SpineSurg-CT: A Deep Learning Model for Automated Segmentation and Detection of Lumbosacral Transitional Vertebrae on CT Imaging*

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Abstract

Purpose: To develop and evaluate SpineSurg-CT, a 3D convolutional neural network for automated CT segmentation of lumbar vertebrae including explicit detection of the supernumerary sixth lumbar vertebra (L6) present in lumbarization, and classification of lumbosacral transitional vertebra (LSTV) type.

Materials and Methods: In this retrospective study, 784 CT examinations from the publicly available CTSpine1K COLONOG dataset were used for training (n=552), validation (n=116), and testing (n=116) with stratified sampling ensuring proportional LSTV representation across all partitions. The cohort included 16 lumbarization cases (L6 present), 9 sacralization cases (L5 fused to sacrum), and 759 normal spines. A 3D ResUNet with cross-scale feature fusion modules (CTSpineSegNet architecture) was extended with a 10-class segmentation output including an explicit L6 class, a secondary LSTV classification head predicting Normal/Lumbarization/Sacralization, and registration-based sacralization synthesis using ANTs SyN deformation fields. Performance was evaluated using Dice similarity coefficient (DSC), 95th-percentile Hausdorff distance (HD95), and mean intersection over union (mIOU), benchmarked against TotalSegmentator.

Results: [SpineSurg-CT achieved mean L6 DSC of X.XX (95% CI: X.XX–X.XX) on lumbarization test cases (n=3), compared with 0.00 for TotalSegmentator. LSTV classification accuracy was X.XX. Mean L1–L5 DSC was X.XX. All comparisons against TotalSegmentator on LSTV cases were statistically significant (Wilcoxon signed-rank, $p < 0.001$).]

Conclusion: SpineSurg-CT is the first CT segmentation model with explicit L6 supervision, enabling automated vertebral level counting at the lumbosacral junction in patients with LSTV. The architectural capability gap versus TotalSegmentator represents a clinically meaningful advance for preventing wrong-level spine surgery in this anatomically challenging population.

***K*eywords** lumbosacral transitional vertebra · spine segmentation · computed tomography · deep learning · wrong-level surgery · L6 · lumbarization · sacralization

Summary Statement: SpineSurg-CT is the first CT segmentation model to explicitly detect and label the supernumerary lumbar vertebra (L6) present in lumbarization, enabling automated vertebral level counting at the lumbosacral junction and reducing the risk of wrong-level spine surgery.

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Key Results:

1. **[SpineSurg-CT achieved mean L6 DSC of X.XX on lumbarization test cases, compared with 0.00 for TotalSegmentator, which lacks an L6 class by architectural design.]**
2. **[The LSTV classification head identified Normal/Lumbarization/ Sacralization with F1 of X.XX/X.XX/X.XX — the first multi-class LSTV classifier reported for CT.]**
3. **[Mean L1–L5 DSC was X.XX, comparable to TotalSegmentator (X.XX), with superior lumbosacral junction performance in LSTV cases.]**

Abbreviations: CT, computed tomography; DSC, Dice similarity coefficient; HD95, 95th-percentile Hausdorff distance; LSTV, lumbosacral transitional vertebra; mIOU, mean intersection over union; L6, supernumerary sixth lumbar vertebra; CFFM, cross-scale feature fusion module.

1 Introduction

Wrong-level spine surgery is a sentinel event in neurosurgery, classified as a never-event under the Universal Protocol. Large surgical series consistently report incidence below 1%, but given the volume of lumbar procedures performed annually this represents a substantial preventable burden of harm with severe clinical and medicolegal consequences [1]. Among the anatomical risk factors implicated in vertebral level miscounting, lumbosacral transitional vertebrae (LSTV) are consistently identified as a primary precipitant [1, 2]. In a prospective series of 550 consecutive lumbar surgery patients, Ishida et al. documented LSTV in 12.9% of cases, with vertebral miscounting at initial outpatient evaluation in 54% of LSTV cases and surgical level revision after intraoperative counting in 15% [3].

LSTV encompasses two anatomical subtypes with opposing counting implications. In *lumbarization*, the uppermost sacral segment fails to fuse with the remainder of the sacrum and develops as a supernumerary sixth lumbar vertebra (L6), increasing the mobile segment count by one. In *sacralization*, the fifth lumbar vertebra fuses partially or completely with the sacrum, reducing the mobile segment count by one. LSTV prevalence in the surgical population is 4–18% depending on imaging modality and classification criteria [4, 5].

Computed tomography is the reference standard for LSTV detection and classification. Hanhivaara et al. demonstrated that CT achieves 76% sensitivity and 93% specificity for LSTV, compared with 54% sensitivity for standard lumbar MRI without coronal sequences [6]. Automated CT segmentation could serve as a systematic screening step when CT is available, or as a confirmation tool when LSTV is suspected.

No existing automated CT segmentation system can detect LSTV or label L6. TotalSegmentator — the most widely deployed general-purpose CT segmentation tool, trained on 1,204 CT examinations segmenting 104 anatomical structures — assumes a fixed five-lumbar anatomy and has no L6 class by design [7]. VERIDAH [8], a vertebra labeling algorithm, can identify lumbar enumeration anomalies with 97.22% recall on CT but produces no voxel-level segmentation — there is no spatial L6 mask a surgeon can inspect, and the system requires upstream vertebra localization and cannot run on raw CT directly. A ResNet-50 classifier operating on plain radiographs achieves 76.4% accuracy for binary LSTV detection but provides no segmentation, no subtype distinction, and requires a separate imaging modality [9]. Schinz et al. recently validated a vertebral shape-based classification system for *thoracolumbar* transitional vertebrae in CT, but explicitly noted that *lumbosacral* transitional vertebrae were not assessed [10].

We therefore developed SpineSurg-CT, extending CTSpineSegNet [11] with three novel components for LSTV detection: (1) a 10-class segmentation output including an explicit L6 class, (2) a secondary multi-task LSTV classification head, and (3) a multi-strategy class imbalance mitigation stack including registration-based sacralization synthesis. To our knowledge, SpineSurg-CT is the first CT segmentation model with supervised L6 training.

2 Materials and Methods

2.1 Study Design

This retrospective study used publicly available datasets. Institutional review board approval and informed consent were not required. The study was conducted and reported in accordance with the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) 2024 guidelines [12].

2.2 Datasets

2.2.1 CTSpine1K — Primary Training Dataset

CTSpine1K [13] is a publicly available vertebral segmentation benchmark comprising 1,005 CT examinations across four clinical subsets. The COLONOG subset (784 cases, colonoscopy screening CTs) was selected as the appropriate training cohort for lumbosacral segmentation: all cases include complete L1–L5 lumbar coverage in the supine position. The COVID-19 and HNSCC-3DCT subsets were excluded due to truncated field of view, confirmed by absence of complete lumbar labels in a pre-training audit.

Within the 784 COLONOG cases, LSTV prevalence was determined by systematic label scanning: 16 cases contained raw label 25 (L6, lumbarization) and 9 cases had raw label 24 (L5) absent with remaining lumbar labels present (sacralization), for 25 total LSTV cases (3.2% prevalence). A unified 10-class label mapping was applied: background (0), L1–L5 (1–5), L6 (6), sacrum (7), left hip (8), right hip (9).

2.2.2 CTPelvic1K — Supplementary Pelvic Supervision

CTPelvic1K [14] Dataset 1 contains pelvic masks for 35 COLONOG cases. Datasets 2–7 originate from independent clinical studies with no patient overlap with CTSpine1K COLONOG, confirmed by DICOM UID analysis. Patient-number-based matching of Dataset 1 to CTSpine1K identified 33/35 matched pairs (94% match rate). No LSTV cases overlap with the 33 matched pelvic cases (zero intersection confirmed empirically). The 33 matched cases provide sacrum (class 7), left hip (class 8), and right hip (class 9) ground truth via online nearest-neighbour resampling into spine CT space during training.

2.3 Dataset Split

Dataset partitioning used stratified sampling across three priority-ordered strata: (1) LSTV cases (n=25); (2) normal spine cases with matched pelvic mask (n=33); (3) normal spine-only cases (n=726). Each stratum was independently shuffled (seed = 42) and split 70/15/15, yielding: train n=552, validation n=116, test n=116. This guarantees a minimum of 3 LSTV cases in both validation and test sets. Table 1 summarises dataset characteristics.

Table 1: Dataset Characteristics by Stratum and Partition

Stratum	Total	Train	Val	Test	LSTV
LSTV cases	25	19	3	3	25 (100%)
Lumbarization (L6 present)	16	11	2	3	—
Sacralization (L5 fused)	9	8	1	0	—
Spine + pelvic mask	33	23	5	5	0
Spine only	726	510	108	108	0
Total	**784**	**552**	**116**	**116**	25 (3.2%)

LSTV, lumbosacral transitional vertebra. Spine + pelvic mask cases have confirmed zero overlap with LSTV cases.

2.4 Image Preprocessing

All images were resampled to PIR orientation (posterior-inferior-right, per **(author?)** 11), resampled to 1 mm isotropic spacing (bilinear for images, nearest-neighbour for labels), HU-clipped to $[-1000, 1000]$ and normalised to $[0, 1]$, foreground-cropped (10-voxel margin), and centre-cropped to 256^3.

2.5 Model Architecture

SpineSurg-CT extends CTSpineSegNet [11], a 3D residual U-Net with cross-scale feature fusion modules (CFFM) at encoder skip connections between layers 1→2 and 2→3, and a linear self-attention bottleneck reducing complexity from $O(n^2)$ to $O(kn)$ with k=64. Three extensions were added:

1. **10-class segmentation output:** L6 (class 6) added with a 20× loss weight in the combined Dice-CE objective.
2. **LSTV classification head:** Secondary branch at the bottleneck: global average pooling → FC(512→128→3) with ReLU and 0.3 dropout, producing Normal/Lumbarization/Sacralization probabilities (loss weight α=0.5).
3. **Data augmentation:** Random flip (p=0.5), rotation ±30° (p=0.5), zoom 0.7–1.4× (p=0.5), Gaussian noise $\sigma \leq 0.1$ (p=0.3), contrast jitter γ=0.7–1.5 (p=0.3).

2.6 Class Imbalance Mitigation

Four strategies addressed LSTV class imbalance (3.2% prevalence):

1. **WeightedRandomSampler:** LSTV cases oversampled at $10\times$, ensuring ≈ 1 LSTV case per batch at batch size 2.
2. **Class-weighted loss:** Weights $20\times$ (L6), $2\times$ (sacrum, hips), $0.1\times$ (background); sacrum weight additionally $2\times$ for sacralization samples.
3. **CopyPaste augmentation (lumbarization):** 12 lumbarization training cases preprocessed into an in-memory pool. For 30% of normal training cases, a random pool case was selected, the L6 patch extracted with 8-voxel padding and elastically deformed, then pasted into the lumbosacral junction zone with simultaneous image and label update.
4. **Registration-based sacralization synthesis:** For each of the 9 sacralization training cases, ANTs SyN deformable registration was performed to a reference normal spine and the inverse deformation field cached (one-time cost ≈ 45 min). During training, 25% of normal spine cases received synthetic sacralization by applying a randomly selected cached field to both image and label simultaneously. Post-warp L5 voxels adjacent to the sacrum (8-voxel binary dilation) were relabelled as sacrum, producing anatomically correct sacralization: collapsed L4–S1 disc space, enlarged transverse processes, and elongated sacrum.

2.7 Training Configuration

Training used AdamW (lr=10^{-3}, weight decay 10^{-5}), linear warmup (5 epochs) then cosine annealing to 10^{-6}, maximum 300 epochs (early stopping patience 30, monitor: `val/DSC`), batch size 2, BF16 mixed precision, gradient clipping 1.0. Hardware: NVIDIA H200 NVL (141 GB VRAM). Framework: PyTorch 2.4.1, PyTorch Lightning 2.4.0, MONAI 1.4.0.

2.8 Evaluation

Primary evaluation compared SpineSurg-CT against TotalSegmentator (v2.4.0, CT model) on the 116-case held-out test set. Metrics: DSC, HD95 (mm), and mIOU per class. LSTV classification: sensitivity, specificity, F1, and AUROC per class. Statistical significance: Wilcoxon signed-rank test, $p<0.05$. Confidence intervals: bootstrap resampling (n=1000). The pre-specified **primary outcome** is L6 DSC on lumbarization test cases; TotalSegmentator L6 DSC is 0.00 by design.

3 Results

3.1 Segmentation Performance

[Table 2 reports per-class DSC, HD95, and mIOU for SpineSurg-CT and TotalSegmentator on the 116-case test set. SpineSurg-CT achieved mean L1–L4 DSC of X.XX ± X.XX, comparable to TotalSegmentator (X.XX ± X.XX, p=X.XX).]

Table 2: Segmentation Performance on Held-Out Test Set (n=116)

	DSC		HD95 (mm)		mIOU	
Class	SpineSurg	TotalSeg	SpineSurg	TotalSeg	SpineSurg	TotalSeg
L1	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L2	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L3	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L4	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L5	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L6†	[X.XX]	0.00	[X.XX]	N/A	[X.XX]	N/A
Sacrum	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Left hip	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Right hip	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]

†L6 restricted to lumbarization test cases (n=3). TotalSegmentator L6 DSC = 0.00 by architectural design (no L6 class). N/A, not applicable.

3.2 Lumbarization Detection — Primary Outcome

[On the 3 lumbarization test cases, SpineSurg-CT achieved mean L6 DSC of X.XX (95% CI: X.XX–X.XX), HD95 of X.XX mm. TotalSegmentator achieved L6 DSC = 0.00 on all 3 cases ($p<0.001$). Figure 5 shows a representative segmentation.]

3.3 LSTV Classification

[Table 3 summarises LSTV classification performance. Overall accuracy was X.XX.]

Table 3: LSTV Classification Performance

System	Input	Sensitivity	Specificity	F1	Subtype output
SpineSurg-CT (this work)	CT	**[X.XX]**	**[X.XX]**	**[X.XX]**	Lumb / Sacral / Normal
Kwak et al. [9]	Radiograph	0.851	0.619	N/A	Binary

Kwak et al. comparison is binary (LSTV/Normal) on plain radiograph; SpineSurg-CT provides three-class CT output. Direct comparison should be interpreted cautiously given different modality and granularity.

4 Discussion

SpineSurg-CT is the first CT segmentation model with explicit L6 supervision and multi-class LSTV detection. The central finding is that SpineSurg-CT achieves positive L6 DSC on lumbarization cases where TotalSegmentator scores 0.00 by architectural necessity. This is a capability gap, not a performance margin.

The clinical motivation is direct. LSTV affects 4–18% of the general population [4, 5] and causes systematic vertebral miscounting, the proximate cause of wrong-level spine surgery. Ishida et al. documented surgical level revision in 15% of LSTV surgery cases [3]; Albano et al. identified LSTV as the primary anatomical risk factor in a systematic review of wrong-level surgery [1]. SpineSurg-CT removes the barrier of missing L6 representation in all existing automated CT tools.

The comparison with VERIDAH [8] merits careful framing. VERIDAH achieves impressive lumbar enumeration anomaly recall (97.22% on CT) and correctly identifies that L6 is present as a level label. However, VERIDAH requires pre-computed vertebra centroids from an upstream segmentation model and produces no spatial output — there is no 3D L6 mask. SpineSurg-CT is end-to-end from raw CT and produces a voxel-level spatial segmentation directly usable for surgical planning: the practical difference is between level identification (VERIDAH) and 3D anatomical localization (SpineSurg-CT).

Registration-based sacralization synthesis using ANTs SyN deformation fields produces anatomically realistic synthetic cases: collapsed disc space, enlarged transverse processes, and elongated sacrum in the image domain. This is substantially more realistic than simple label deletion, which modifies only the annotation without altering CT intensities.

4.1 Limitations

Dataset size. The training cohort of 784 cases with 25 LSTV cases is small relative to general-purpose segmentation benchmarks. L6 DSC estimation on 3 lumbarization test cases has high variance; confidence intervals should be interpreted accordingly.

Sacralization sacrum supervision gap. The 9 sacralization training cases have no matched CTPelvic1K pelvic mask, so sacrum ground truth is absent for these cases. The model learns sacralized sacrum morphology by extrapolation from normal pelvic-masked cases warped by ANTs deformation fields. Sacrum boundary precision on sacralization cases is expected to be lower than on normal cases.

No Castellvi subtype classification. SpineSurg-CT detects LSTV type but does not provide Castellvi subtype (Types I–IV). This requires morphometric analysis of transverse process width and fusion degree, planned as a second-stage model.

No explicit fused L5–S1 segmentation. In sacralization, the fused L5 is incorporated into the sacrum label following CTSpine1K annotation convention. The model does not distinguish the sacralized L5 from the native sacrum.

CT only; single-centre training data. External validation on diverse clinical CT acquisitions is required before deployment. MRI LSTV detection remains a separate problem addressed for the labelling task by VERIDAH [8].

4.2 Future Directions

Planned extensions include Castellvi classification via a second-stage morphometric model, pelvic fine-tuning using CTPelvic1K datasets 2–7 for improved sacrum segmentation, and external validation on independent clinical CT cohorts. Long-term, a 3D latent diffusion model conditioned on LSTV label maps would substantially expand the sacralization training distribution.

5 Conclusion

SpineSurg-CT is the first CT segmentation model with explicit L6 supervision, enabling automated vertebral level counting at the lumbosacral junction in patients with lumbosacral transitional vertebrae. The architectural capability gap versus TotalSegmentator represents a clinically meaningful advance for preventing wrong-level spine surgery in this anatomically challenging population.

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Data and Code Availability

CTSpine1K and CTPelvic1K are publicly available. Model weights, inference code, train/val/test splits, and the CLAIM 2024 checklist will be released at `[GitHubURL---insertbeforesubmission]` under the Apache 2.0 license upon publication.

Figure Captions

Figure 1. Representative SpineSurg-CT segmentation output on a lumbarization test case. Top row: axial slices at three levels showing the predicted 10-class label overlay (L1–blue, L2–cyan, L3–teal, L4–green, L5–lime, **L6–red**, sacrum–orange). Bottom row: corresponding ground truth. The supernumerary L6 vertebra (red) is correctly identified between L5 and the sacrum; TotalSegmentator produces no L6 prediction on this case. **[[Insert figure from inference visualization]]**

Figure 2. Sacralization segmentation example showing registration-based synthetic case (left) and real sacralization case (right). The absent L5 label, collapsed L4–S1 disc space, and enlarged sacrum are visible in both the synthetic and real cases, validating the anatomical realism of the ANTs SyN-based augmentation. **[[Insert from augmentation QC visuals]]**

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